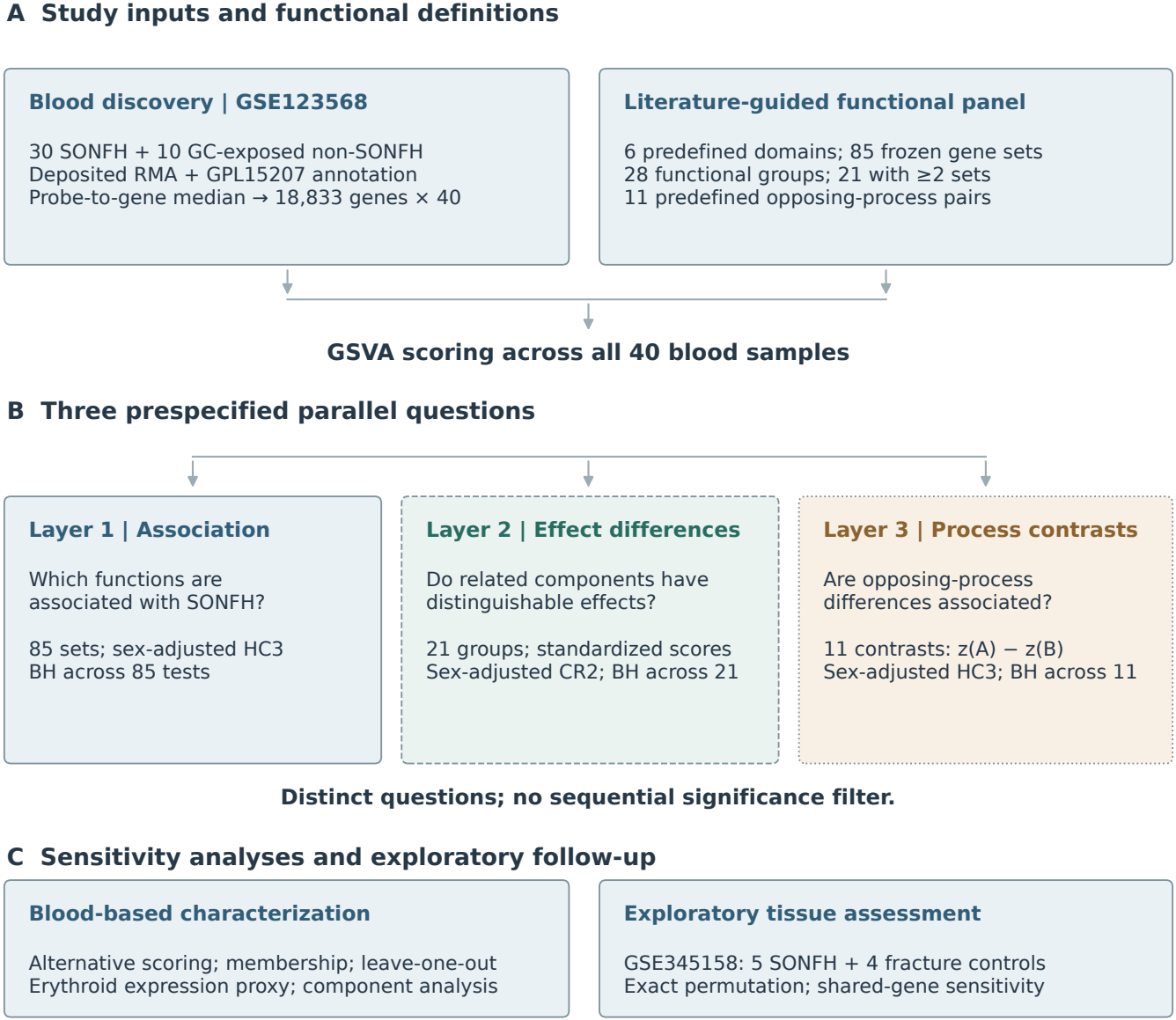
A Study inputs and functional definitions
Blood discovery | GSE123568
30 SONFH + 10 GC-exposed non-SONFH
Deposited RMA + GPL15207 annotation
Probe-to-gene median → 18,833 genes × 40
Literature-guided functional panel
6 predefined domains; 85 frozen gene sets
28 functional groups; 21 with ≥2 sets
11 predefined opposing-process pairs
GSVA scoring across all 40 blood samples
B Three prespecified parallel questions
Layer 1 | Association
Which functions are associated with SONFH?
85 sets; sex-adjusted HC3
BH across 85 tests
Layer 2 | Effect differences
Do related components have distinguishable effects?
21 groups; standardized scores
Sex-adjusted CR2; BH across 21
Layer 3 | Process contrasts
Are opposing-process differences associated?
11 contrasts: z(A) − z(B)
Sex-adjusted HC3; BH across 11
Distinct questions; no sequential significance filter.
C Sensitivity analyses and exploratory follow-up
Blood-based characterization
Alternative scoring; membership; leave-one-out
Erythroid expression proxy; component analysis
Exploratory tissue assessment
GSE345158: 5 SONFH + 4 fracture controls
Exact permutation; shared-gene sensitivity